Quiz — 2.1.4 Thinking Logically — ANSWER SHEET

Separate answer sheet / mark scheme — keep for marking.

Answer key

Section A (6 marks — 1 each) 1. **B** — a decision point is where flow of control branches on a condition. 2. **B** — "under 16" means strictly less than 16, so `age < 16`. 3. **B** — when the condition is false there are not enough seats, so the booking is rejected. 4. **C** — a single IF/ELSE decision has two outcomes (ON and OFF). 5. **B** — a loop's repeat-condition is a decision point tested each pass, steering flow of control. 6. **B** — locking on the third failure means `attempts >= 3`.

Section B (9 marks)

7. **[3]** Condition (1): `IF score >= 1000`. TRUE branch (1): award the player a bonus life. FALSE branch (1): do not award a life / continue play unchanged. Must give the exact `>=` Boolean *and* both outcomes.

8. **[2]** Condition (1): `IF ticketValid = TRUE AND feePaid = TRUE` (accept `IF ticketValid AND feePaid`). False outcome (1): the barrier stays closed / the driver is refused exit and prompted to pay. Must use a logical operator (AND).

9. **[2]** Flow of control = the order in which a program's statements are executed (1). Decision points alter that order by branching to different statements depending on whether a condition is true or false — e.g. `IF balance >= price` runs the "buy" path, otherwise the "decline" path (1).

10. **[2]** Under `total > 50` a £50 order does **not** qualify, because 50 is not greater than 50 (1). To include exactly £50, use `IF total >= 50` (1).

Section C (5 marks) 11. **[5]** Up to 5 from: First/overriding decision on connection speed — `IF connectionSpeed < 5` (Mbps) THEN stream SD, regardless of subscription (1 for condition, 1 for both/overriding outcome). Otherwise (connection adequate) decide on subscription — `IF subscription = "premium"` (or `isPremium = TRUE`) THEN stream 4K, ELSE stream HD (1 for condition, 1 for both branch outcomes). Recognise the conditions are nested/ordered so the speed test takes priority (1). Conditions must be exact Booleans with both branches stated and applied to this scenario; vague "if it's fast" wording does not earn the condition marks.

Total: 6 + 9 + 5 = 20